

Program to print details of a Car from Class Vehicle:

```
helloworld.cpp x a.exe
CODES > helloworld.cpp > ...
1 #include <iostream>
2 using namespace std;
3
4 class Vehicle {
5 private:
6     string registration_number;
7     int year;
8 public:
9     Vehicle(string reg, int y) {
10         registration_number = reg;
11         year = y;
12     }
13     void printDetails() {
14         cout << "Registration Number: " << registration_number << endl;
15         cout << "Year: " << year << endl;
16     }
17 };
18
19 class Car : public Vehicle {
20 private:
21     string model;
22 public:
23     Car(string reg, int y, string m) : Vehicle(reg, y) {
24         model = m;
25     }
26     void printDetails() {
27         Vehicle::printDetails();
28         cout << "Model: " << model << endl;
29     }
30 };
31
32 int main() {
33     // Create a Car object and print its details
34     Car c1("ABC123", 2020, "Honda Civic");
35     c1.printDetails();
36
37     return 0;
38 }
39
```

```
PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Registration Number: ABC123
Year: 2020
Model: Honda Civic
PS D:\CODES>
```

Program to create class Vehicle and print details of a car from it:

```
helloworld.cpp x a.exe
CODES > helloworld.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     const int SIZE = 5;
6     double prices[SIZE];
7     double total = 0, average;
8
9     cout << "Enter " << SIZE << " prices:\n";
10    for (int i = 0; i < SIZE; i++) {
11        cout << "Price " << i+1 << ": ";
12        cin >> prices[i];
13        total += prices[i];
14    }
15
16    average = total / SIZE;
17    cout << "Average price: " << average << endl;
18
19    cout << "Prices higher than the average:\n";
20    for (int i = 0; i < SIZE; i++) {
21        if (prices[i] > average) {
22            cout << prices[i] << endl;
23        }
24    }
25
26    return 0;
27 }
```

```
PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Enter 5 prices:
Price 1: 75
Price 2: 55
Price 3: 60
Price 4: 30
Price 5: 80
Average price: 60
Prices higher than the average:
75
80
PS D:\CODES>
```

Program to calculate average of prices and those who are higher than average:

```
helloworld.cpp x a.exe
CODES > helloworld.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     const int SIZE = 5;
6     double prices[SIZE];
7     double total = 0, average;
8
9     cout << "Enter " << SIZE << " prices:\n";
10    for (int i = 0; i < SIZE; i++) {
11        cout << "Price " << i+1 << ": ";
12        cin >> prices[i];
13        total += prices[i];
14    }
15
16    average = total / SIZE;
17    cout << "Average price: " << average << endl;
18
19    cout << "Prices higher than the average:\n";
20    for (int i = 0; i < SIZE; i++) {
21        if (prices[i] > average) {
22            cout << prices[i] << endl;
23        }
24    }
25
26    return 0;
27 }
```

```
PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Enter 5 prices:
Price 1: 75
Price 2: 55
Price 3: 60
Price 4: 30
Price 5: 80
Average price: 60
Prices higher than the average:
75
80
PS D:\CODES>
```

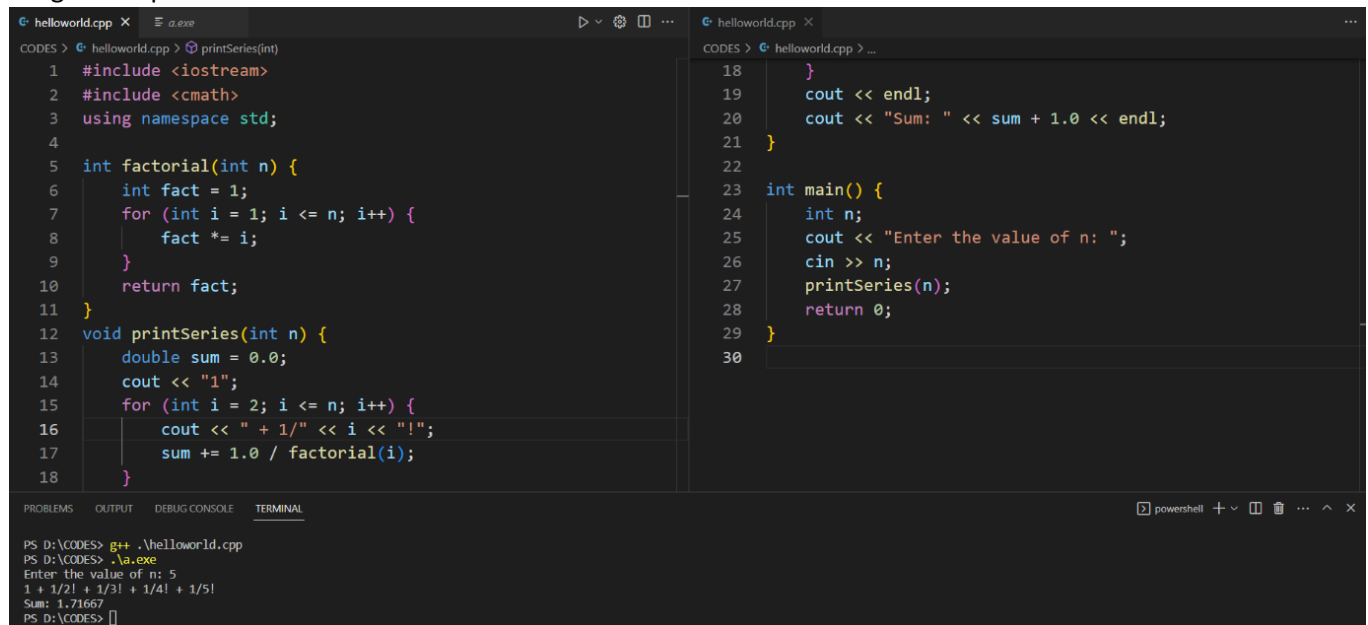
Program to perform various operations on a string:

```
helloworld.cpp x a.exe
CODES > helloworld.cpp > main()
1 #include <iostream>
2 #include <cstring>
3 #include <cctype>
4 using namespace std;
5
6 int main() {
7     char str[100];
8     char ch;
9     int length, frequency = 0, i;
10    bool isPalindrome = true;
11
12    cout << "Enter a string: ";
13    cin.getline(str, 100);
14
15    length = strlen(str);
16    cout << "Length of the string: " << length << endl;
17
18    cout << "Enter a character to find its frequency: ";
19    cin >> ch;
20    for (i = 0; i < length; i++) {
21        if (str[i] == ch) {
22            frequency++;
23        }
24    }
25    cout << "Frequency of " << ch << " in the string: "
26    << frequency << endl;
27 }
```

```
helloworld.cpp x
CODES > helloworld.cpp > main()
26 << frequency << endl;
27
28 for (i = 0; i < length; i++) {
29     if (isupper(str[i])) {
30         cout << str[i] << " is in uppercase." << endl;
31     }
32     else if (islower(str[i])) {
33         cout << str[i] << " is in lowercase." << endl;
34     }
35 }
36
37 for (i = 0; i < length/2; i++) {
38     if (str[i] != str[length-i-1]) {
39         isPalindrome = false;
40         break;
41     }
42 }
43
44 if (isPalindrome) {
45     cout << "The string is a palindrome." << endl;
46 }
47 else {
48     cout << "The string is not a palindrome." << endl;
49 }
50
51 return 0;
52 }
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Enter a string: Computer
Length of the string: 8
Enter a character to find its frequency: e
Frequency of e in the string: 1
C is in uppercase.
o is in lowercase.
m is in lowercase.
p is in lowercase.
u is in lowercase.
t is in lowercase.
e is in lowercase.
r is in lowercase.
The string is not a palindrome.
PS D:\CODES>
```

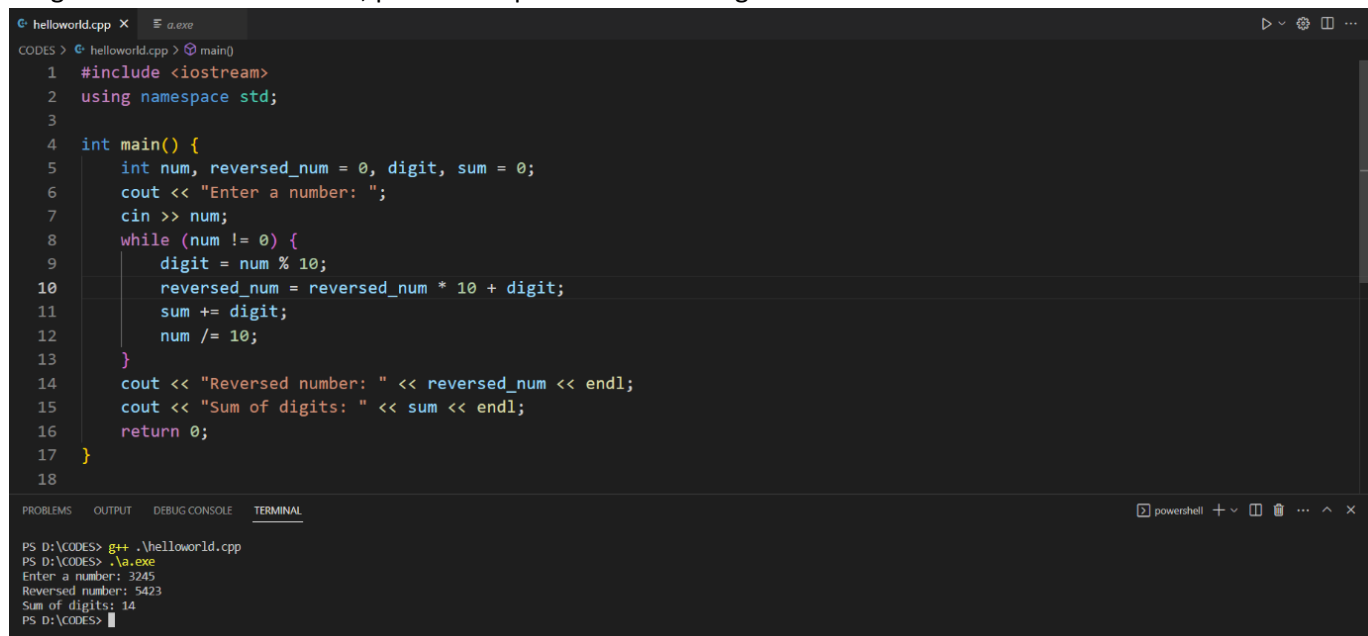
Program to print the series and its sum:



```
helloworld.cpp x a.exe
CODES > helloworld.cpp > printSeries(int)
1  #include <iostream>
2  #include <cmath>
3  using namespace std;
4
5  int factorial(int n) {
6      int fact = 1;
7      for (int i = 1; i <= n; i++) {
8          fact *= i;
9      }
10     return fact;
11 }
12 void printSeries(int n) {
13     double sum = 0.0;
14     cout << "1";
15     for (int i = 2; i <= n; i++) {
16         cout << " + 1/" << i << "!";
17         sum += 1.0 / factorial(i);
18     }
19 }
20
21
22
23 int main() {
24     int n;
25     cout << "Enter the value of n: ";
26     cin >> n;
27     printSeries(n);
28     return 0;
29 }
30

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Enter the value of n: 5
1 + 1/2! + 1/3! + 1/4! + 1/5!
Sum: 1.71667
PS D:\CODES>
```

Program to reverse a number, print it and print the sum of digits:



```
helloworld.cpp x a.exe
CODES > helloworld.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num, reversed_num = 0, digit, sum = 0;
6      cout << "Enter a number: ";
7      cin >> num;
8      while (num != 0) {
9          digit = num % 10;
10         reversed_num = reversed_num * 10 + digit;
11         sum += digit;
12         num /= 10;
13     }
14     cout << "Reversed number: " << reversed_num << endl;
15     cout << "Sum of digits: " << sum << endl;
16     return 0;
17 }
18

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Enter a number: 3245
Reversed number: 5423
Sum of digits: 14
PS D:\CODES>
```

A menu driven program:

The screenshot displays a multi-panel IDE with three code editors and a terminal. The left editor shows the `fibonacci()` function, which generates the Fibonacci series. The middle editor shows the `factorial()` function and the `main()` function, which includes a menu for the user to choose between checking a number's parity, displaying the Fibonacci series, computing a factorial, or exiting. The right editor shows the `main()` function's logic, which uses a `switch` statement to handle the user's choice and a `while` loop to repeat the menu until the user chooses to exit. The terminal at the bottom shows the program's execution, including the compilation command `g++ helloworld.cpp` and the output of the program for various inputs.

```
1 #include <iostream>
2 using namespace std;
3
4 void checkOddEven() {
5     int n;
6     cout << "Enter a number: ";
7     cin >> n;
8     if (n % 2 == 0) {
9         cout << n << " is even." << endl;
10    } else {
11        cout << n << " is odd." << endl;
12    }
13 }
14
15 void fibonacci() {
16     int n, t1 = 0, t2 = 1, nextTerm;
17     cout << "Enter the number of terms: ";
18     cin >> n;
19     cout << "Fibonacci Series: ";
20     for (int i = 1; i <= n; ++i) {
21         cout << t1 << " ";
22         nextTerm = t1 + t2;
23         t1 = t2;
24         t2 = nextTerm;
25     }
26     cout << endl;
27 }
28
29 void factorial() {
30     int n, fact = 1;
31     cout << "Enter a positive integer: ";
32     cin >> n;
33     for (int i = 1; i <= n; i++) {
34         fact *= i;
35     }
36     cout << "Factorial of " << n << " = " << fact << endl;
37 }
38
39 int main() {
40     int choice;
41     do {
42         cout << "Choose an option:" << endl;
43         cout << "1. Check if a number is odd or even" << endl;
44         cout << "2. Display the Fibonacci series" << endl;
45         cout << "3. Compute the factorial of a number" << endl;
46         cout << "4. Exit" << endl;
47         cout << "Enter your choice: ";
48         cin >> choice;
49         switch (choice) {
50             case 1:
51                 checkOddEven();
52                 break;
53             case 2:
54                 fibonacci();
55                 break;
56             case 3:
57                 factorial();
58                 break;
59             case 4:
60                 cout << "Exiting program..." << endl;
61                 break;
62             default:
63                 cout << "Invalid choice. Try again." << endl;
64         }
65     } while (choice != 4);
66     return 0;
67 }
```

PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Choose an option:
1. Check if a number is odd or even
2. Display the Fibonacci series
3. Compute the factorial of a number
4. Exit
Enter your choice: 1
Enter a number: 889
889 is odd.
Choose an option:
1. Check if a number is odd or even
2. Display the Fibonacci series
3. Compute the factorial of a number
4. Exit
Enter your choice: 2
Enter the number of terms: 3
Fibonacci Series: 0 1 1
Choose an option:
1. Check if a number is odd or even
2. Display the Fibonacci series
3. Compute the factorial of a number
4. Exit
Enter your choice: 3
Enter a positive integer: 4
Factorial of 4 = 24
Choose an option:
1. Check if a number is odd or even
2. Display the Fibonacci series
3. Compute the factorial of a number
4. Exit
Enter your choice: 4
Exiting program...
PS D:\CODES>

Program to take n number and print the lines with '*':

The screenshot shows an IDE with a single code editor and a terminal. The code editor contains a C++ program that takes a positive integer `n` as input and prints a pattern of asterisks. The pattern consists of `n` lines, where the `i`-th line contains `i` asterisks. The terminal shows the program's execution for the input `5`, resulting in a 5x5 grid of asterisks.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cout << "Enter a positive integer: ";
7     cin >> n;
8     for (int i = 1; i <= n; i++) {
9         for (int j = 1; j <= i; j++) {
10             cout << "* ";
11         }
12         cout << endl;
13     }
14     return 0;
15 }
16
```

PS D:\CODES> g++ .\helloworld.cpp
PS D:\CODES> .\a.exe
Enter a positive integer: 5
*
* *
* * *
* * * *
* * * * *
PS D:\CODES>

